

Lab 3: Component Modelling & Architectural Pattern Selection — Written Justification

Name: Dinesh Ashok Jadage | SRN: PES1UG25CS813 | PES University, B.Tech CSE

Problem Statement #63 — Sustainability & Green Tech: Community Solar Credit Allocation Manager

Architecture Selection: We chose Layered Architecture for the Community Solar Credit Allocation Manager System.

1. Requirements and Challenges

Functional (from Lab 1): FR-001 record solar generation from smart meters; FR-002 transfer surplus credits between residents; FR-003 show credit balance; FR-004 compute monthly bill offset; FR-005 generation, transfer and billing reports for the Co-op Manager. **Non-functional:** NFR-001 up to 500 households with 99.9% uptime; NFR-002 encrypted user data and credit transactions. **Challenges:** consistent credit balances (no double-spend), availability, secure handling of financial-style data, and a simple portal for two user roles.

2. Comparison of Architectural Styles

Style	Pros for this system	Cons for this system	Verdict
Layered	Portal / services / ledger split matches the requirements; one consistent ledger; simple to secure and operate	Layers are coupled; scales by replicating the whole business tier	Selected
Microservices	Independent scaling and fault isolation	Credit transfer touches two balances, so distributed transactions and data-consistency risk; heavy operations for only 500 households	Over-engineered
Client-Server	Central control, easy data consistency	Single server is a single point of failure, which conflicts with the 99.9% uptime target; no internal structure	Too coarse

3. Components and Interfaces (see component diagram)

Seven components in three layers plus one external system: **Smart Meter Gateway** (external); **Web/Mobile Portal** (Presentation); **Generation Tracker, Credit Manager, Billing Engine, Report Service** (Business); **Credit Ledger Database** (Data). Interfaces: **ISmartMeter** (MQTT/HTTPS over TLS), **ICreditService** and **IReportService** (REST over HTTPS), **ICreditBalance** (Billing Engine → Credit Manager, API call) and **ILedger** (SQL over TLS, used by the four business components).

4. Justification

Reason 1 — Credit integrity needs one consistent ledger. A transfer (FR-002) must debit one resident and credit another atomically, and the bill offset (FR-004) must use the same balances. Layered architecture keeps all of this on a single ledger with ordinary database transactions, avoiding the distributed-consistency problems that microservices would add.

Reason 2 — Right-sized and modular for 500 households. The requirements split naturally into portal, business services and data, so each FR lands in one component. Because services talk only through interfaces, the meter vendor, database or billing rules can be replaced without touching the Portal, and the load (about 500 hourly readings) does not justify microservice overhead.

Security Advantage (NFR-002). The Portal never touches the ledger; all requests pass through the Business Layer, which checks the role (Resident or Co-op Manager) and validates the sender's balance before any transfer. Meter, portal and database links use TLS and the ledger is encrypted at rest, so credits and user data are protected end to end at a small number of controlled entry points.

Performance / Availability Benefit (NFR-001). Business components are stateless, so two or more instances can run behind a load balancer to reach 99.9% uptime. Calls between layers are direct, and indexed ledger queries keep balance checks and transfers fast; monthly reports read stored records rather than recomputing raw readings.

5. Trade-off Acknowledged

Layers are coupled and scale together. For a single neighbourhood microgrid of about 500 households this is acceptable in return for consistency, simplicity and easier security.